

 Pattern	 Real-Life Analogy	 Solves
Sliding Window	"Peeking through a moving window"	Longest Substring, Max Sum Subarray, Anagrams
Two Pointers	"2 fingers walking toward each other"	2Sum, Reverse Vowels, Sorted Square, Palindrome Check
Fast & Slow Pointers	"Tom & Jerry: one fast, one slow" 🐭🐱	Linked List Cycle, Happy Number
Binary Search	"Guess the number in 7 tries or less" 🎯	Rotated Array, First/Last Position, Koko Eating Bananas
DFS / BFS	"DFS: Go deep, BFS: Go wide"	Graph Traversals, Flood Fill, Word Ladder, Shortest Path
Backtracking	"Trying all outfits before choosing one" 👗	Sudoku, N-Queens, Word Search, Permutations
Dynamic Programming	"Why re-solve what you've already solved?"	Knapsack, House Robber, LIS, Edit Distance
Greedy	"Always pick what seems best right now"	Activity Selection, Jump Game, Gas Station
Union-Find (DSU)	"Friend circles detection" 🧑🧑	Number of Provinces, Kruskal's MST, Connected Components
Topological Sort	"Finish A before B" 🧱	Course Schedule, Task Scheduling
Prefix Sum / Difference Array	"Running totals like bank statements" 💵	Range Sum, Subarray Sum Equal K, Rainwater Trapping
Monotonic Stack / Queue	"Stacking plates or tallest first" 🍽️	Next Greater Element, Daily Temperatures, Largest Rectangle
Bit Manipulation	"Flip switches to solve puzzles" 📺	Single Number, Count Bits, Subsets, XOR Problems
Trie (Prefix Tree)	"Autocomplete dictionary" 📖	Word Search, StartsWith, Replace Words
Heap / Priority Queue	"Serve most urgent first" ☕	Kth Largest, Top K Elements, Merge K Lists
Graph (Adjacency List/Matrix)	"Follow paths on a map" 🗺️	Shortest Path, Cycles, Connected Components
Recursion	"Function calling itself – like Russian dolls" 🎉	Tree Problems, Factorial, Subsets, Merge Sort
Segment Tree / Fenwick Tree	"Smart range queries" 📊	Range Sum/Min/Max, Updates on Array

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Matrix Traversal	“Walk in all directions” 	Spiral Order, Island Count, Diagonal Traversals
Hashing / HashMap	“Lookup table memory” 	2Sum, Group Anagrams, Frequency Counts